

Project Charter — P08 Stock Market Sentiment and Price Movement Predictor

Project Overview

This project builds an integrated AI system that predicts next-day stock price direction by combining a deep learning time-series model with a financial sentiment analysis pipeline. The assignment document defines P08 as a FinTech and algorithmic trading project with a deep learning objective based on an LSTM time-series model and an NLP objective based on transformer-driven financial sentiment analysis.[cite:1]

The project is designed for moderate difficulty with an estimated effort of 25–30 hours total, and it is mapped in the assignment to ML Engineer, Data Scientist, and NLP Engineer roles. [cite:1] The final system will combine historical OHLCV market data with transformer-based sentiment scores so that the fused model can be compared against a price-only baseline. [cite:1]

Problem Statement

Stock price movement prediction is difficult when using only historical price data because market behavior is influenced by both numerical patterns and information flow from financial news. P08 addresses this by combining technical indicators derived from historical market data with sentiment extracted from financial text, creating a richer predictive signal than either modality alone.[cite:1]

For an aspiring data-oriented engineer, this project is valuable because it demonstrates ingestion of multiple data sources, feature engineering, sequence modeling, sentiment modeling, and evaluation through backtesting. The assignment also explicitly expects comparison experiments, honest reporting of metrics, and reproducible implementation practices, which align well with placement-ready portfolio work.[cite:1]

Project Objectives

Deep Learning Objective

The deep learning objective is to build a stacked LSTM model for next-day price direction classification using five years of OHLCV data for five stocks. The assignment specifies engineering input features such as returns, 20-day rolling standard deviation, RSI, and MACD, using a 30-day lookback sequence, dropout between LSTM layers, Adam optimization, and a backtesting comparison against Buy-and-Hold on one year of out-of-sample data.[cite:1]

NLP Objective

The NLP objective is to build a financial news sentiment classifier using a transformer model such as ProsusAI/FinBERT. The assignment requires tokenization and POS analysis, three-class sentiment classification, fusion of sentiment with the LSTM price features, and presentation of a correlation heatmap between sentiment score and one-day price movement.[cite:1]

Integrated Objective

The unified project goal is to build a complete pipeline in which structured market features and unstructured sentiment features are fused into a single predictive system. The assignment explicitly requires comparison of a price-only LSTM model against an LSTM-plus-sentiment model and expects reporting of the resulting accuracy improvement.[cite:1]

Scope

In Scope

- Historical OHLCV data collection for five stocks using a free public source such as yfinance.[cite:1]
- Feature engineering on financial time-series data, including returns, rolling volatility, RSI, and MACD.[cite:1]
- Development of a stacked LSTM model for binary direction prediction.[cite:1]
- Financial sentiment classification using a transformer model such as FinBERT.[cite:1]
- Fusion of sentiment scores with price-derived features for final prediction.[cite:1]
- Backtesting and metric reporting including accuracy, AUC, and Sharpe ratio.[cite:1]
- A demo interface for showcasing the full workflow in the project presentation.[cite:1]

Out of Scope

- Real-time trading execution or broker integration.
- Intraday or high-frequency forecasting.
- Portfolio optimization and risk allocation across positions.
- Use of paid financial APIs or proprietary datasets, since the assignment requires publicly available datasets only.[cite:1]

Deliverables

The assignment requires each group to submit a working code repository, a 2–3 page technical report, and a 10-minute demo or viva presentation.[cite:1] In practical terms, the project should therefore produce the following deliverables:

Deliverable	Description
Code repository	Reproducible source code with notebooks or scripts for both objectives, clearly labeled, along with a requirements.txt file.[cite:1]
Technical report	A concise 2–3 page report covering approach, experiments, and results for both objectives.[cite:1]
Results summary table	A combined results table reporting evaluation metrics for the deep learning and NLP objectives.[cite:1]
Demo application	A lightweight frontend, such as a Streamlit dashboard, for presentation and viva demonstration.
README	A recruiter-friendly project overview including problem, approach, results, and demo visuals, reflecting the assignment's placement tips.[cite:1]

Stakeholders and Roles

The primary stakeholders are the two student team members, the course evaluators, and potential recruiters or interviewers who may later review the GitHub repository. The assignment specifies teams of exactly two students and emphasizes individual contribution during the presentation and viva.[cite:1]

A practical role split for this project is shown below.

Team Member	Primary Responsibility
Member 1	Data ingestion, feature engineering, LSTM modeling, backtesting
Member 2	Sentiment pipeline, transformer fine-tuning or adaptation, POS analysis, dashboard integration
Both	Fusion logic, evaluation, report writing, presentation, and viva preparation

Functional Requirements

The system should satisfy the following functional requirements derived directly from the assignment brief.

- The system shall collect five years of OHLCV data for five selected stocks.[cite:1]
- The system shall compute engineered financial features including returns, rolling standard deviation, RSI, and MACD.[cite:1]
- The system shall train a stacked LSTM model on 30-day lookback sequences for binary next-day direction prediction.[cite:1]
- The system shall perform sentiment classification on financial text into positive, negative, and neutral classes.[cite:1]
- The system shall aggregate or align sentiment features with market data at a daily level for fusion modeling.[cite:1]
- The system shall compare a price-only model with a fused price-plus-sentiment model and report the performance difference.[cite:1]

- The system shall generate evaluation artifacts including accuracy, AUC, Sharpe ratio, and a sentiment-price correlation heatmap.[cite:1]

Non-Functional Requirements

The assignment includes several practical constraints and placement-oriented expectations that shape the non-functional requirements.

- The implementation must be reproducible, with dependencies clearly specified in a requirements.txt file.[cite:1]
- The repository must be understandable and presentation-ready, with a clear README because recruiters judge repositories quickly.[cite:1]
- The code must avoid plagiarism and any AI-generated code must be disclosed and understood by the team.[cite:1]
- The datasets used must be publicly available and should not be committed to the repository if they exceed 500 MB.[cite:1]
- The model development process should include comparative experiments and honest metric reporting rather than inflated unsupported claims.[cite:1]

Technology Stack

The assignment explicitly highlights PyTorch and Hugging Face as widely asked-about tools in AI/ML interviews, and P08 specifically lists PyTorch, Hugging Face FinBERT, yfinance, LSTM, spaCy, and NewsAPI among the project technologies.[cite:1] A practical stack for implementation is shown below.

Layer	Recommended Tools
Data ingestion	yfinance,pandas
Feature engineering	pandas, numpy, technical indicator utilities
Deep learning	PyTorch
NLP and transformers	transformers,ProsusAI/finbert
NLP preprocessing	spaCy,tokenizer utilities
Evaluation	scikit-learn,matplotlib,seaborn
Demo frontend	Streamlit,Plotly or Matplotlib
Version control	GitHub

Risks and Mitigation

Risk	Impact	Mitigation
Small or noisy sentiment dataset	Medium	Use a well-known public financial sentiment dataset and document limitations clearly.

Risk	Impact	Mitigation
Overfitting in the LSTM model	Medium	Use dropout, validation monitoring, and careful train-test separation based on time.
Data leakage in time-series experiments	High	Split chronologically and keep the final year strictly out-of-sample for backtesting.
Limited compute availability	Medium	Use Google Colab or another accessible GPU environment for transformer runs.
Weak demo impact	Medium	Build a simple Streamlit frontend to present predictions, charts, and comparison metrics clearly.

Success Criteria

The project will be considered successful if it satisfies both the academic brief and portfolio-quality expectations. The assignment expects reporting of classification accuracy, AUC, Sharpe ratio, and comparative gains from adding sentiment to the LSTM model.[cite:1]

Success criteria include:

- Completion of both the deep learning and NLP objectives as one integrated system.[cite:1]
- A working and reproducible code repository with clearly separated modules or notebooks.[cite:1]
- A results table comparing baseline and fusion approaches.[cite:1]
- A presentation-ready demo showing the end-to-end pipeline.[cite:1]
- Clear explanation of methodology, limitations, and future work, which the assignment specifically recommends for placement readiness.[cite:1]

Timeline

A compact two-week execution plan fits the effort level mentioned in the assignment.[cite:1]

Phase	Duration	Tasks
Planning	Day 1-2	Project charter, SRS, architecture diagram, data schema, experiment plan
Data pipeline	Day 3-4	Download market data, clean data, engineer indicators, create labels
LSTM modeling	Day 5-6	Build, train, validate, and evaluate the price-only model
Sentiment pipeline	Day 7-8	Prepare dataset, run transformer sentiment classification, create daily scores
Fusion and backtest	Day 9-10	Join datasets, train fusion model, compare with baseline, backtest
Demo and reporting	Day 11-14	Build Streamlit app, prepare report, update README, finalize presentation

Final Product Vision

The final output should be more than a set of notebooks. A strong final version of this project should present a lightweight dashboard where a user can select a stock, inspect recent price movement, view sentiment score or sentiment class, and see the model's next-day directional prediction along with backtest metrics.

This dashboard-oriented presentation strengthens the project for viva and resume use because it turns the assignment from a model-training exercise into a usable end-to-end analytical system. That aligns well with the assignment's advice to build something worth showing in an interview and to present the work clearly through a README, reproducible code, and comparative results.[cite:1]